



Cambridge IGCSE™

COMPUTER SCIENCE

0478/23

Paper 2

May/June 2020

MARK SCHEME

Maximum Mark: 50

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **11** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)(i)	Any array related to Task 1 – one mark e.g. ItemCode Description Correct purpose related to Task 1 - one mark e.g. ...to store the product code of the items in stock ...to store descriptions/names of the items on sale	2
1(a)(ii)	Any variable related to Task 2 – one mark e.g. Quantity HealthyNum Correct purpose related to Task 2 - one mark e.g. ... to allow input of the quantity of a product ... to store/track the number of healthy items in the order	2
1(a)(iii)	Any constant related to Task 3 – one mark e.g. Off10 Off20 Correct purpose related to Task 3 - one mark e.g. ... to store the 'matching healthy items' option discount rate ... to store the 'all healthy items' discount rate	2
1(b)	Any two correct statements e.g. • The Healthy? data is not numerical • Boolean	2

Question	Answer	Marks
1(c)	Any six from • Input of item/item code (with message) • Attempt at use of validation of input for item/item code • Fully functional validation of input for item/item code • Find location of item/item code in array • Retrieve item description, price and whether it is healthy • Check if another item is required – message and input • Use of loop to repeat purchase option • Running total of price • Output of description of each item, price and healthy status (with message) • Output total price <u>with message</u> (outside final loop) Example answer <pre> AnotherItem ← "Y" While AnotherItem = "Y" SnackCodeFlag ← False OUTPUT "Input an Item Code" WHILE SnackCodeFlag = False INPUT SnackCode Count ← 0 WHILE Count < 22 DO IF SnackCode = ItemCode[Count] THEN SnackCodeFlag ← True OUTPUT "Item Description ", Description[Count], " Item Price ", Price[Count], " Item is Healthy? ", Healthy[Count] TotalPrice ← TotalPrice + Price[Count] Count ← 22 ENDIF Count ← Count + 1 ENDWHILE IF SnackCodeFlag = False THEN </pre>	6

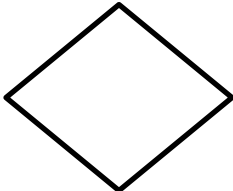
Question	Answer	Marks
1(c)	<pre> OUTPUT "Your Item Code doesn't exist, please try again" ENDIF ENDWHILE OUTPUT "Would you like another item? (Answer Y or N)" INPUT AnotherItem ENDWHILE OUTPUT "The total price is ", TotalPrice </pre>	
1(d)	Any four from • Explanation of comparing numbers of healthy and non-healthy items • Explanation of checking for zero unhealthy items • Explanation of applying the 10% discount rule • Explanation of applying the 20% discount rule • Explanation of calculating the new total price and money saved • Explanation of output	4
1(e)	One for each correct statement (max two) • Use a loop structure when making purchases... / introduce a count of items when making purchases ... /modify the existing loop structure • ... that terminates after six iterations	2

Question	Answer	Marks
2	Description A loop that will iterate at least once A loop that will not be executed on the first test if the condition is false A conditional statement Totalling Counting Pseudocode example CASE ... OF ... OTHERWISE ... ENDCASE Number ← Number + 1 WHILE ... DO ... ENDWHILE Sum ← Sum + NewValue REPEAT ... UNTIL One mark – one correct link Two marks – two correct links Three marks – three correct links Four marks – four/five correct links	4

Question	Answer	Marks
3	Line 2 and Line 4 errors - One mark for each correct identification and correction of error Error 1 line number: Line 2 Correction: REPEAT Error 2 line number: Line 4 Correction: IF Number < 0 OR Number > 499 Line 8 error - One mark for correct identification of error and one mark for each correction of error Error 2 line number: Line 8 Correction: UNTIL Number >= 0 AND Number <= 499	6

Question	Answer	Marks
4(a)	One mark for each correct check (max two) • Length (check) • Type Check • Format Check	2
4(b)	One mark for each suitable piece of test data and one mark for each relevant reason (max four) • LL9999LL999 • Too long • 5678987 • All numeric • CB12EU • No space is present	4

Question	Answer					Marks																																																															
5		<table><tr><th>Flag</th><th>Number</th><th>Divisor</th><th>Value</th><th>OUTPUT</th></tr><tr><td>False</td><td>5</td><td>2</td><td>2</td><td></td></tr><tr><td></td><td></td><td>3</td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td>5 is prime</td></tr><tr><td>False</td><td>6</td><td>2</td><td>3</td><td></td></tr><tr><td>True</td><td></td><td>3</td><td>2</td><td></td></tr><tr><td>True</td><td></td><td>4</td><td></td><td></td></tr><tr><td>False</td><td>8</td><td>2</td><td>4</td><td></td></tr><tr><td>True</td><td></td><td>3</td><td>2</td><td></td></tr><tr><td>True</td><td></td><td>4</td><td>2</td><td></td></tr><tr><td></td><td></td><td>5</td><td></td><td></td></tr><tr><td>False</td><td>0</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table>	Flag	Number	Divisor	Value	OUTPUT	False	5	2	2				3							5 is prime	False	6	2	3		True		3	2		True		4			False	8	2	4		True		3	2		True		4	2				5			False	0										5
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Question	Answer	Marks
6	Decision  Process  One mark for each correct symbol	2

Question	Answer	Marks										
7(a)	<table><tr><th>Fieldname</th><th>Purpose of field contents</th></tr><tr><td>CodeNo</td><td>Primary key to identify products</td></tr><tr><td>Product</td><td>To describe the product</td></tr><tr><td>Price</td><td>The price of individual item</td></tr><tr><td>NumInStock</td><td>How many are in stock</td></tr></table> One mark for each correct fieldname and description pair	Fieldname	Purpose of field contents	CodeNo	Primary key to identify products	Product	To describe the product	Price	The price of individual item	NumInStock	How many are in stock	4
Fieldname	Purpose of field contents											
CodeNo	Primary key to identify products											
Product	To describe the product											
Price	The price of individual item											
NumInStock	How many are in stock											

Question	Answer				Marks
7(b)	Field:	CodeNo	Product	NumInStock	3
	Table:	STOCK	STOCK	STOCK	
	Sort:				
	Show:	☒	☒	☐	
	Criteria:			<20	
	or:				
	One mark for each completely correct column (max three)				